Universitatea POLITEHNICA din București
Facultatea de Automatică și Calculatoare

Electronică analogică

Curs – Material suplimentar 5

Circuite lineare speciale cu amplificatoare operaționale

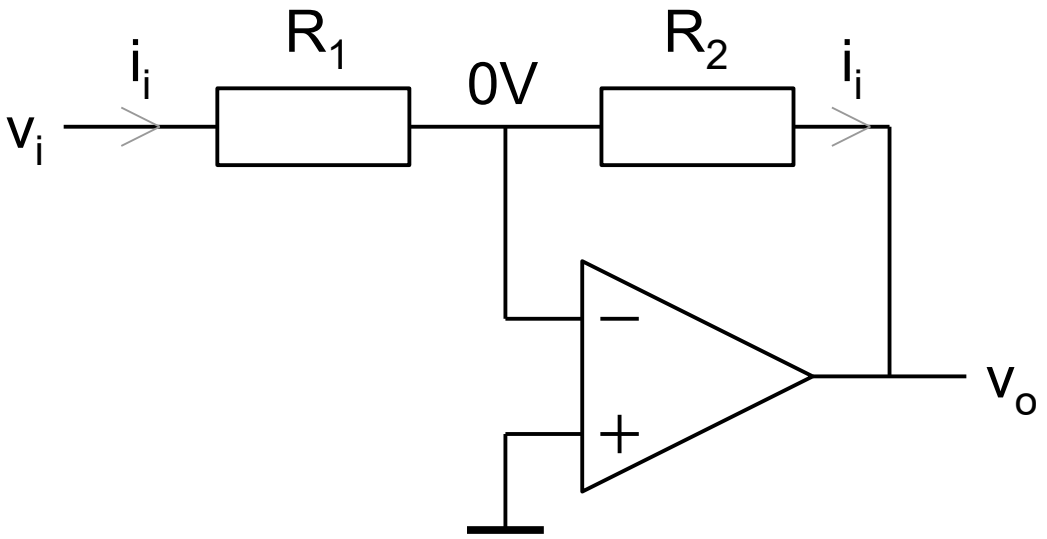
dr. ing. Răzvan Tătăroiu
razvan.tataroiu@upb.ro

2022-2023

Sumar

- Aplicații lineare cu amplificatoare operaționale
  - amplificatoare
    - inversor, neinversor, sumă, diferență
  - filtre
  - integrator, derivator
  - convertor curent-tensiune, tensiune-curent
  - stabilizatoare de tensiune

Inversor



$$i_i = \frac{v_i}{R_1}$$

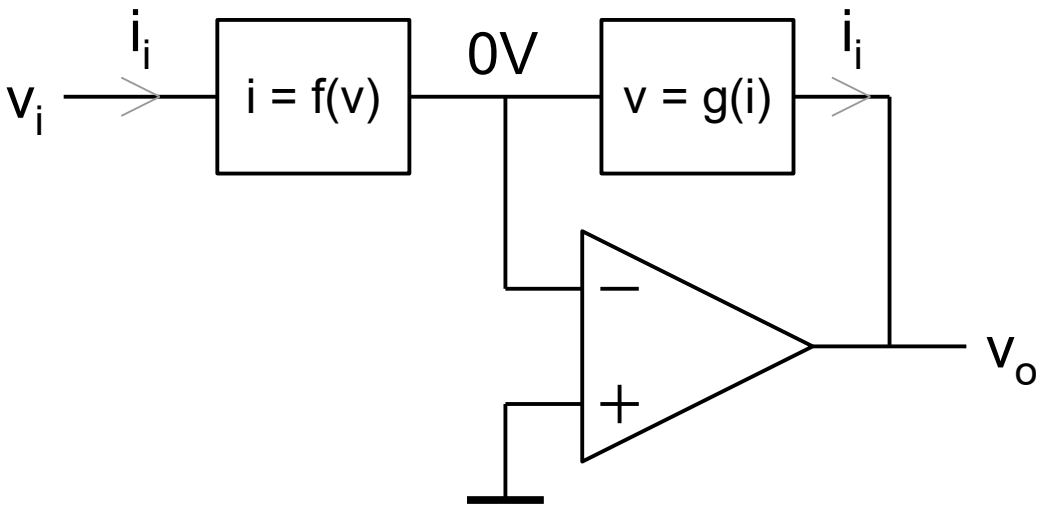
tensiune => curent

$$v_o = -i_i R_2$$

curent => tensiune

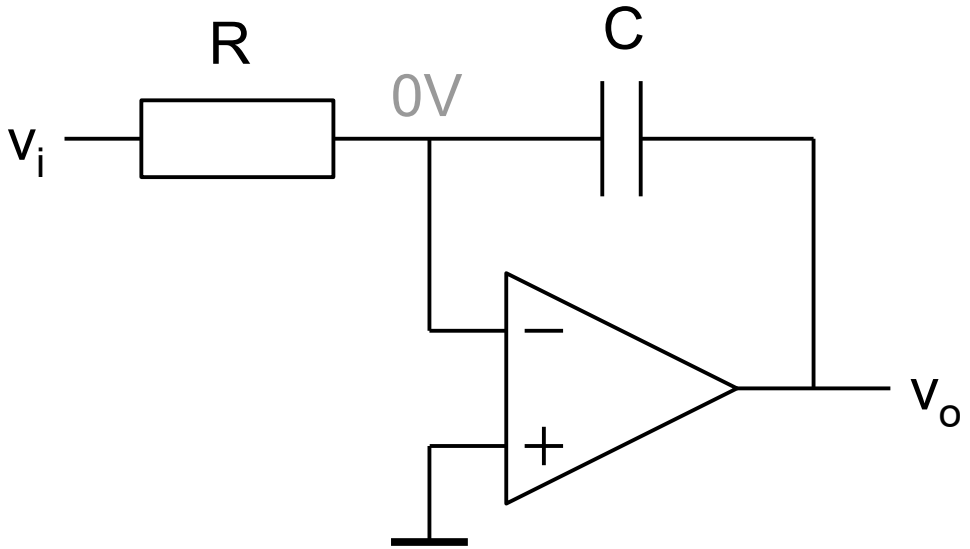
$$v_o = -v_i \frac{R_2}{R_1}$$

Inversor – generalizare



$$v_o = -g\left(f\left(v_i\right)\right)$$

Integrator (1)



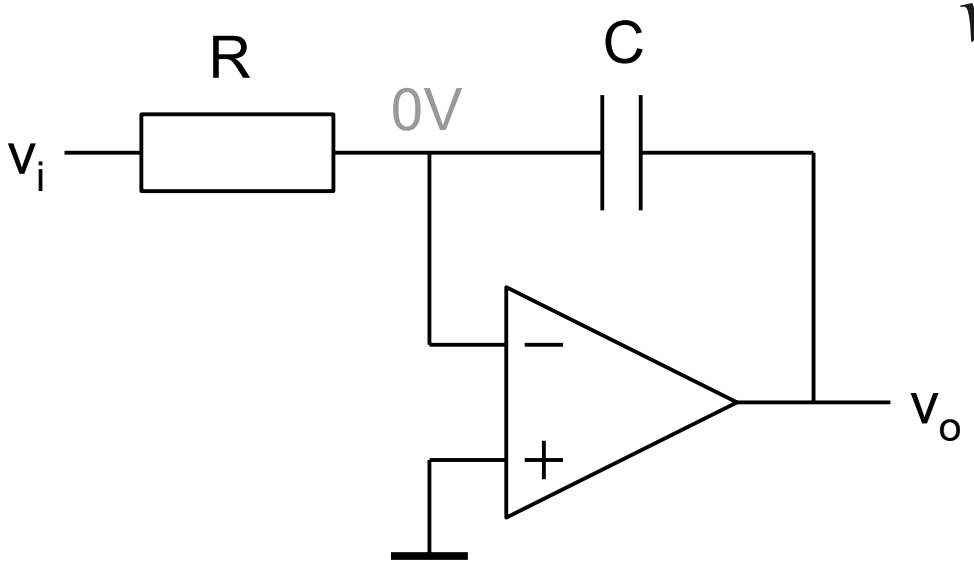
$$i_i = \frac{v_i}{R} = -C\frac{dv_o}{dt}$$

$$\frac{dv_o}{dt} = -\frac{v_i}{RC}$$

$$v_o = -\frac{1}{RC}\int v_i\,dt$$

$$v_o(t) = -\frac{1}{RC}\int_{t_0}^{t} v_i\,d\tau + v_o(t_0)$$

Integrator (2)



$$v_o(t) = -\frac{1}{RC}\int_{t_0}^{t} v_i \, d\tau + v_o(t_0)$$

OBS.:
pentru $v_i = 0$

$$v_o(t) = v_o(t_0)$$

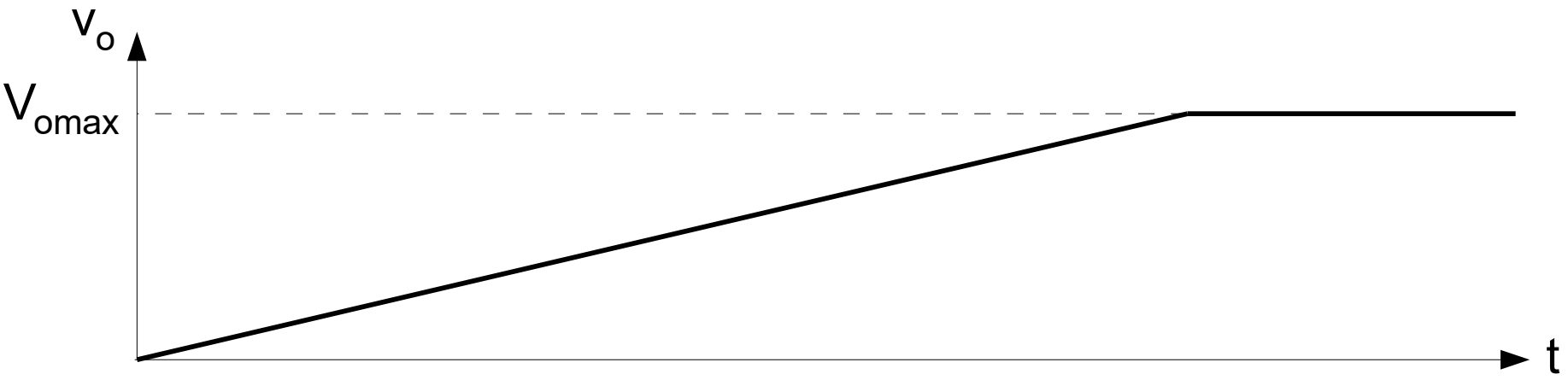
“memorie analogică”

OBS.:
AO prezintă neidealități: tensiune de decalaj V_{OS}, curent de polarizare I_B

Integrator (3)

Contribuțiile neidealităților:

$$v_{o,p} = \frac{1}{RC} \int \left(V_{OS} + I_B R \right) dt$$



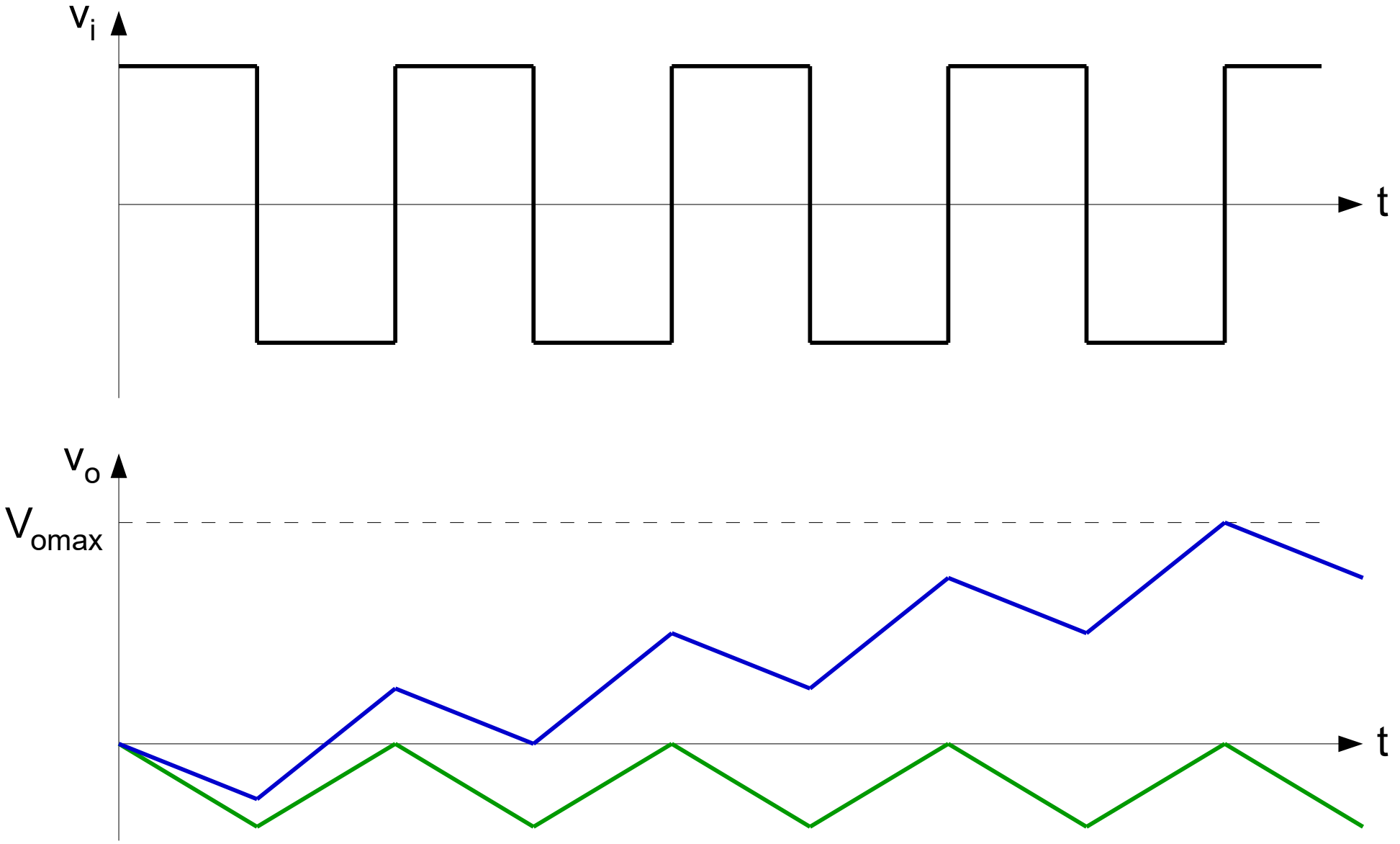
=> rampă parazită

=> la utilizarea pe durate lungi, fenomenul domină ieșirea

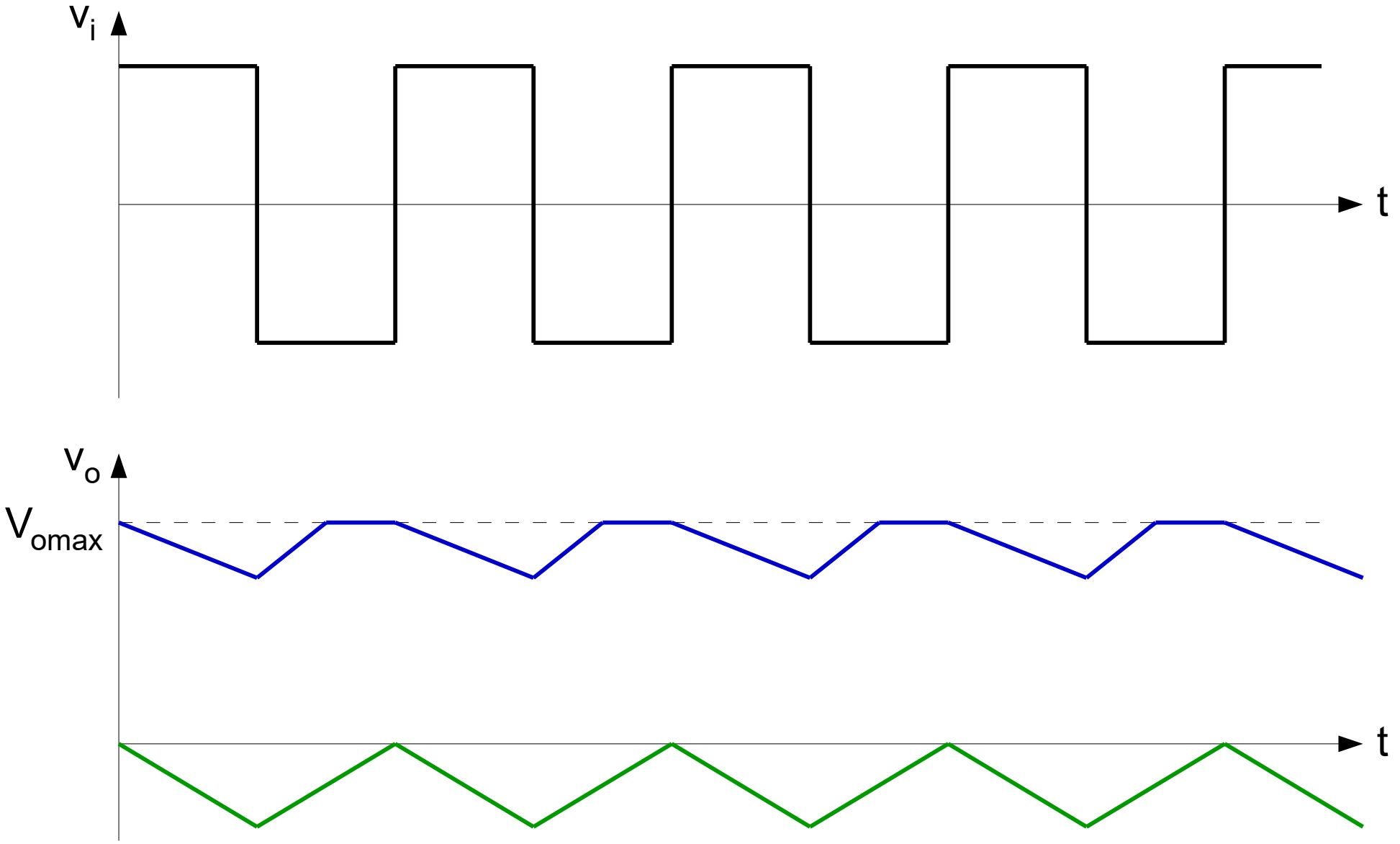
=> ieșirea trebuie menținută în domeniul de funcționare

- resetare periodică

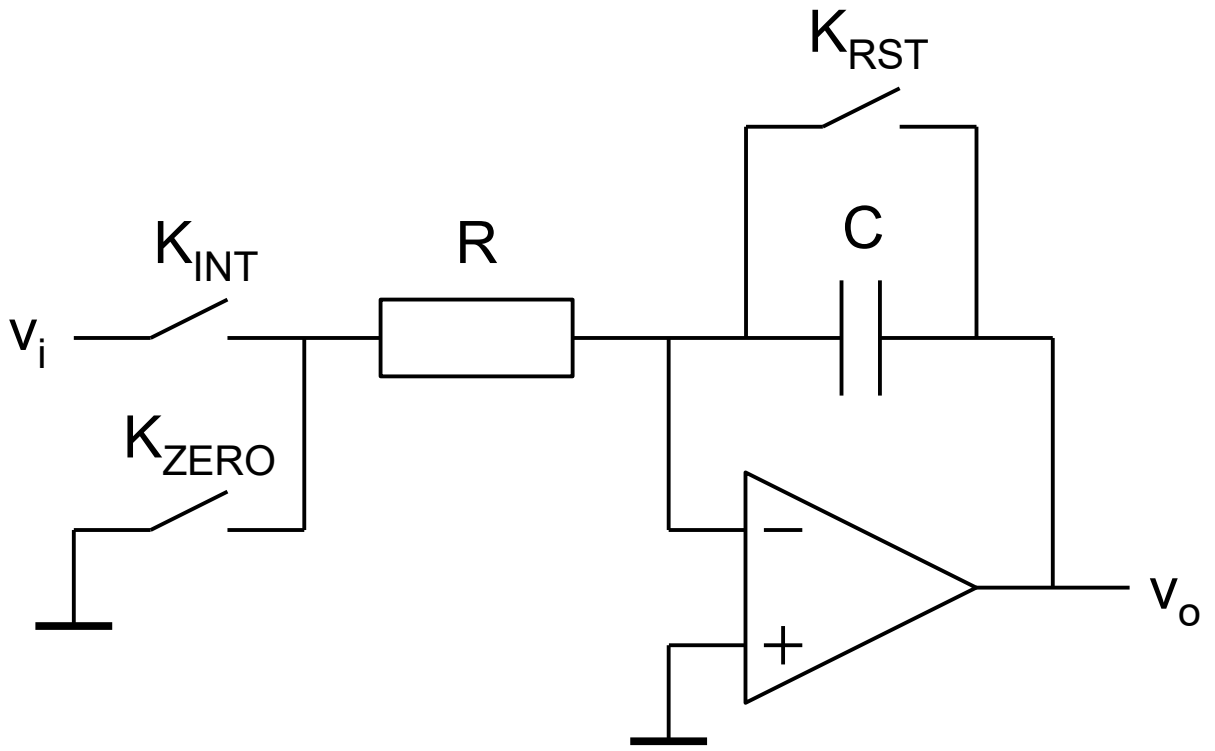
Exemplu – semnal “perfect simetric”



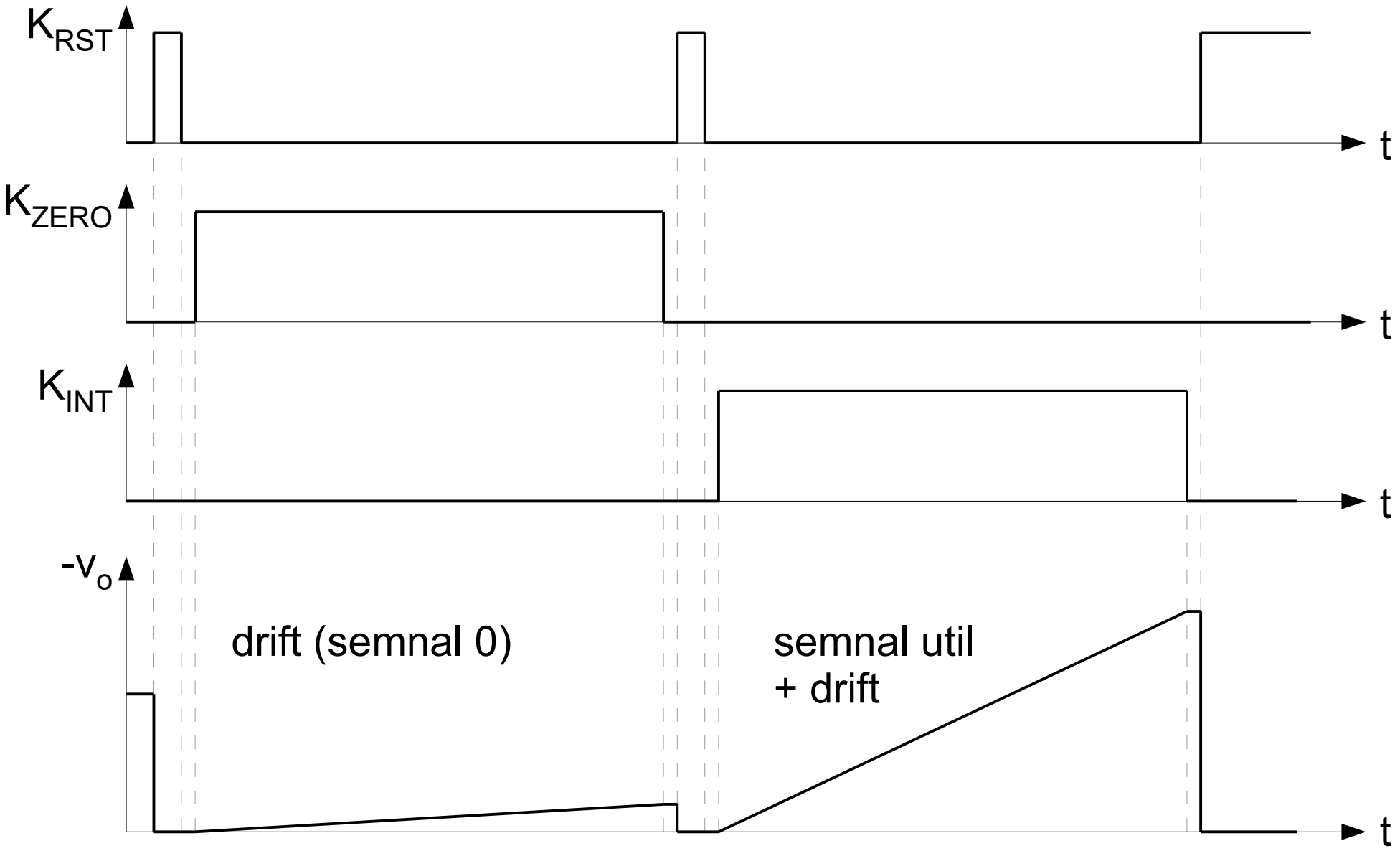
Semnal “perfect simetric” – staționar



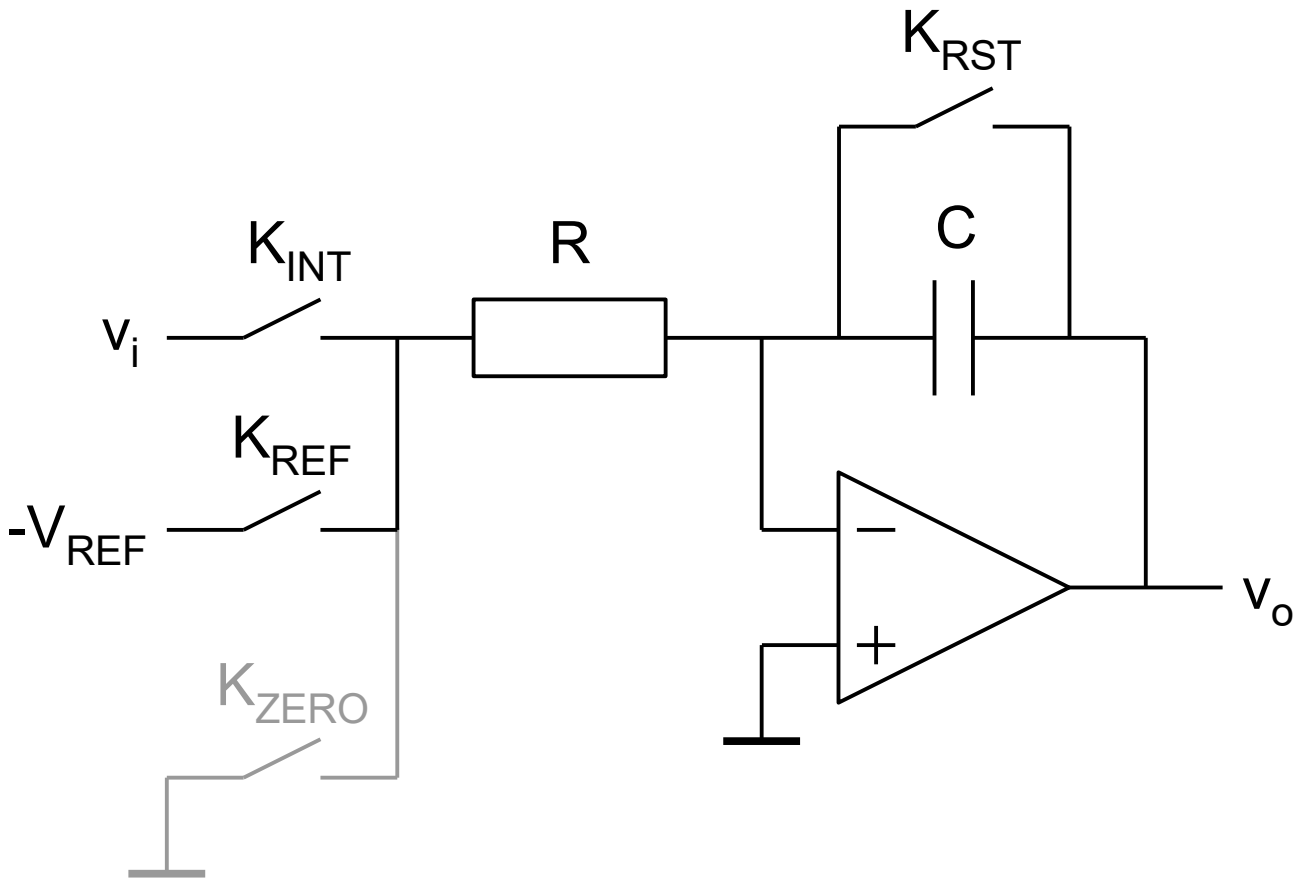
Integrator comutat digital (1)



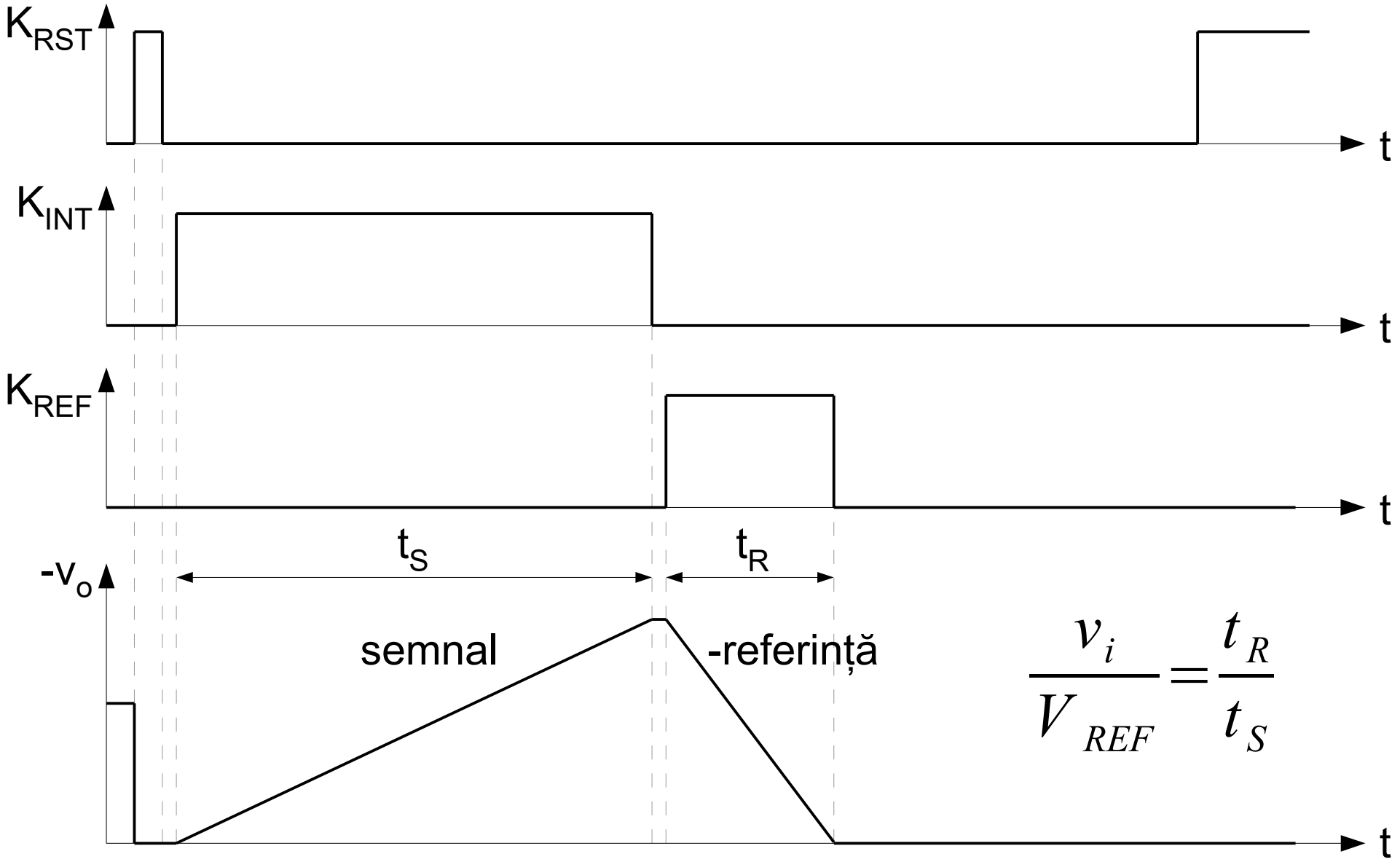
Integrator comutat digital (2)



Convertor A-D cu integrare (1)

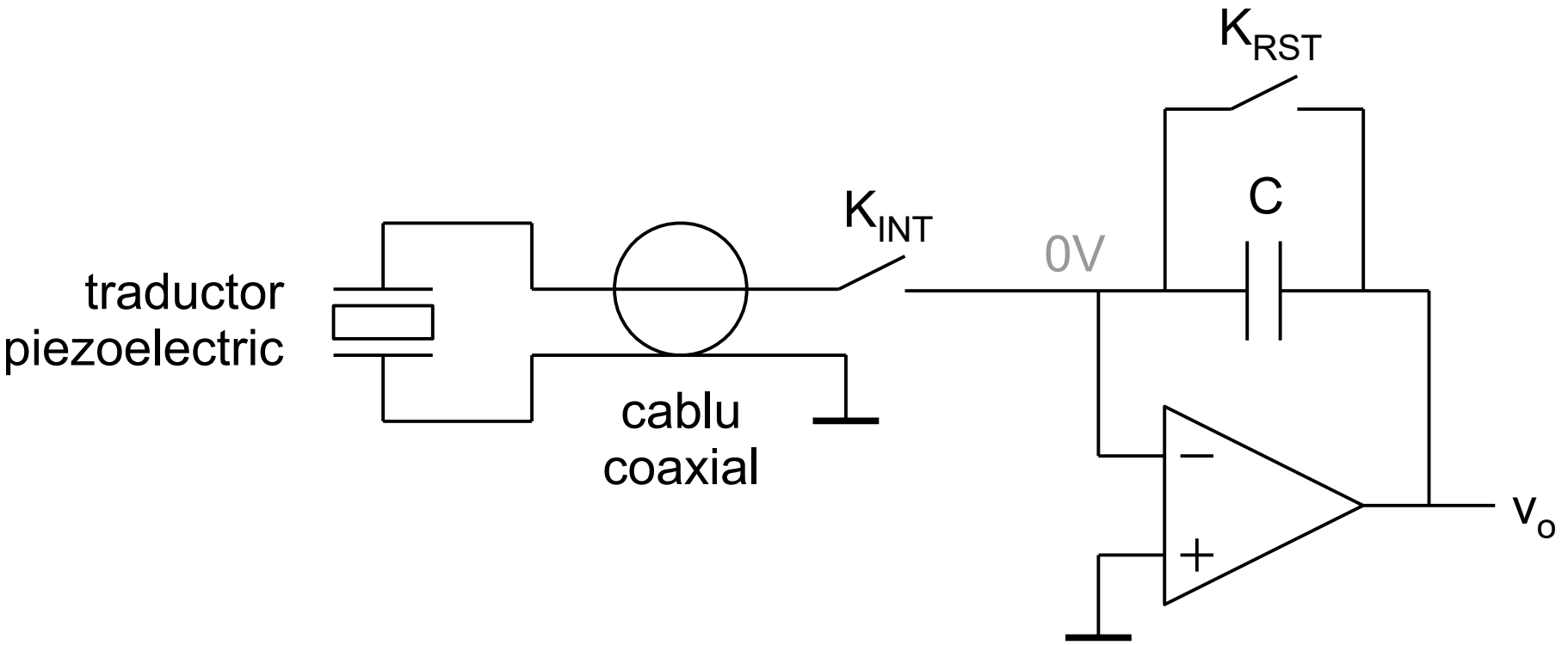


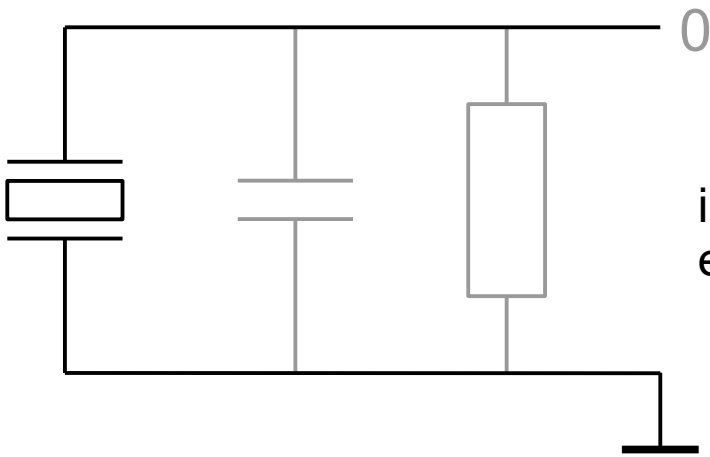
Convertor A-D cu integrare (2)



$$\frac{v_i}{V_{REF}} = \frac{t_R}{t_S}$$

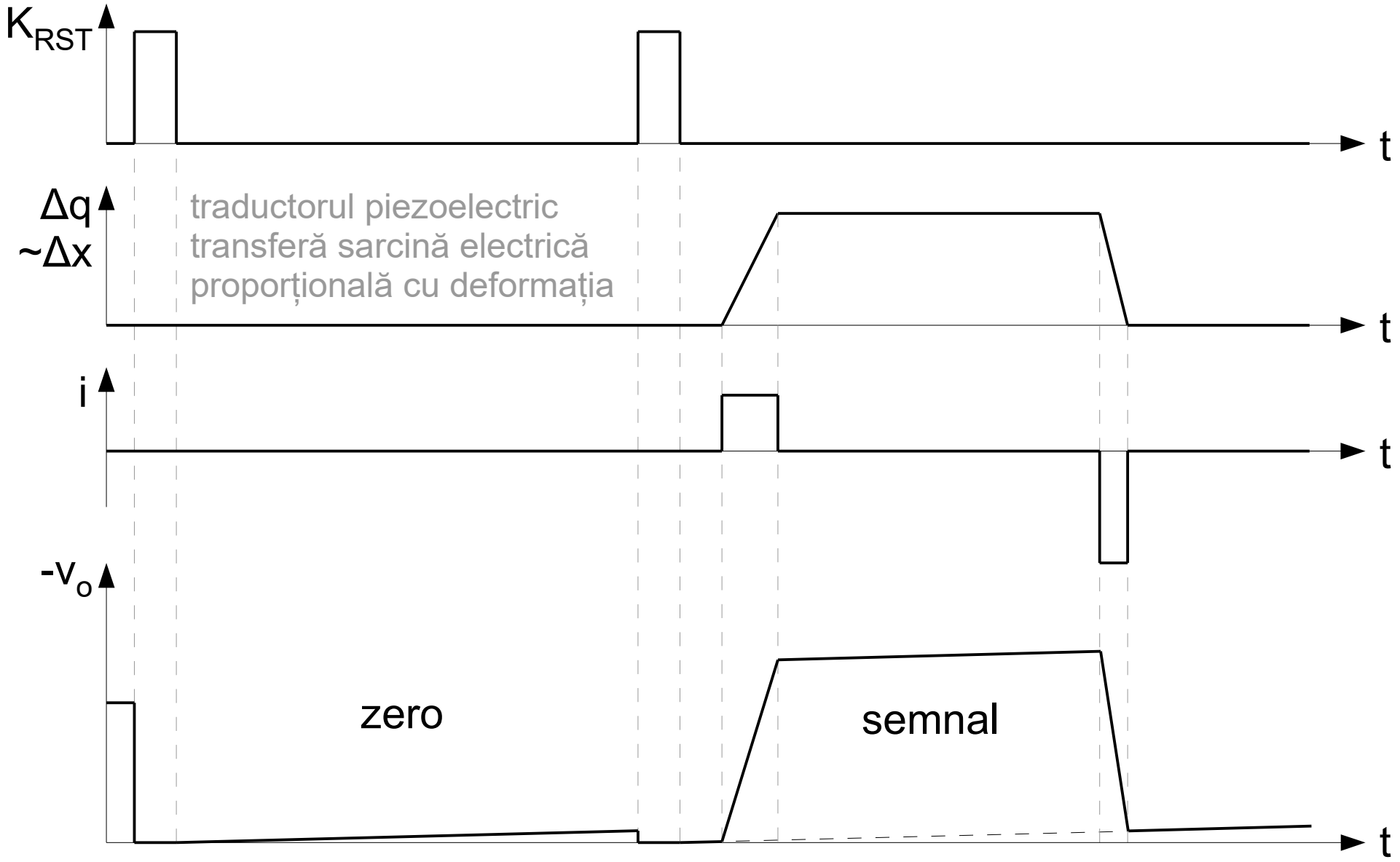
Amplificator de sarcină electrică (1)



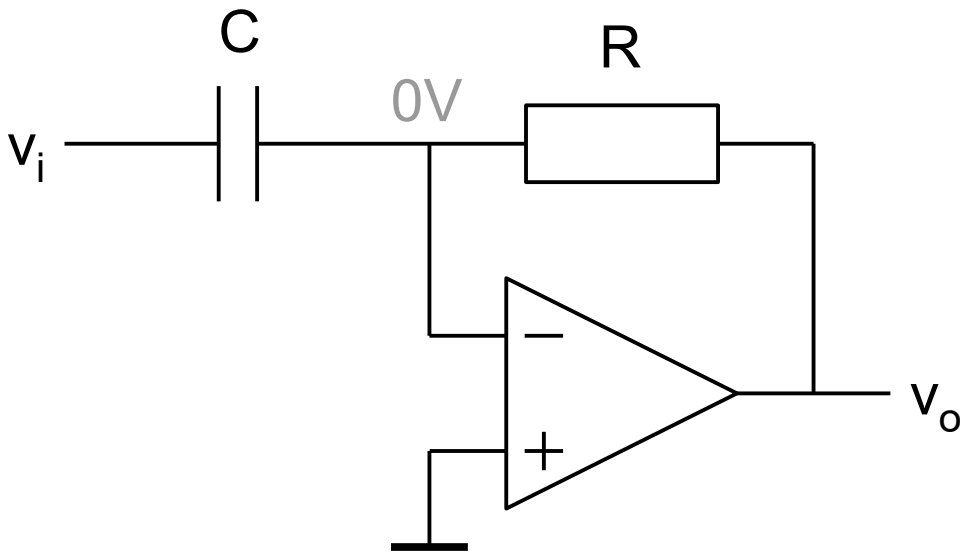


influența elementelor parazite
este redusă grație masei virtuale

Amplificator de sarcină electrică (2)



Derivator (1)



$$i_i = C\frac{dv_i}{dt} = -\frac{v_o}{R}$$

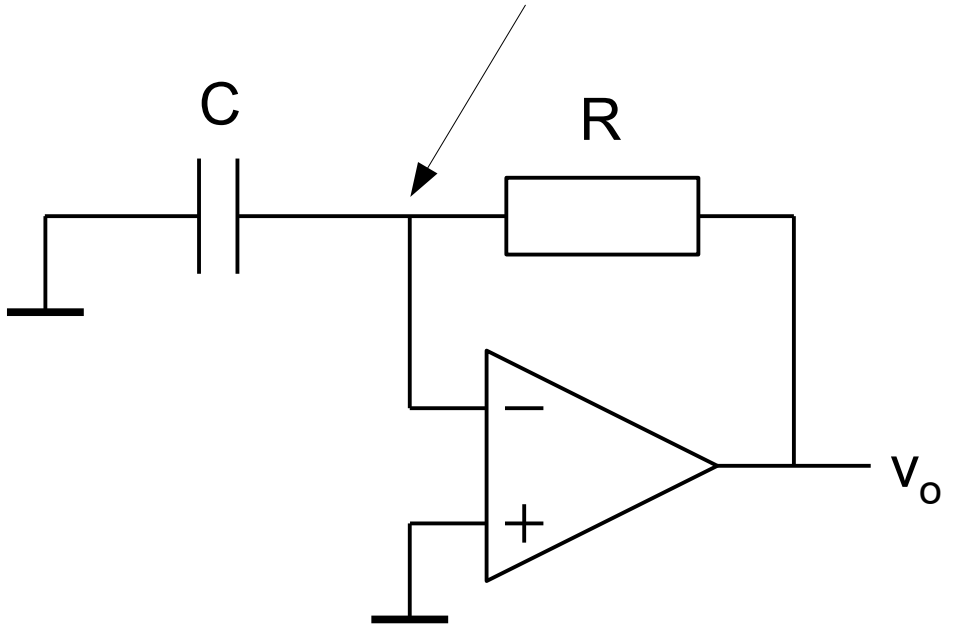
$$v_o = -RC\frac{dv_i}{dt}$$

Probleme:

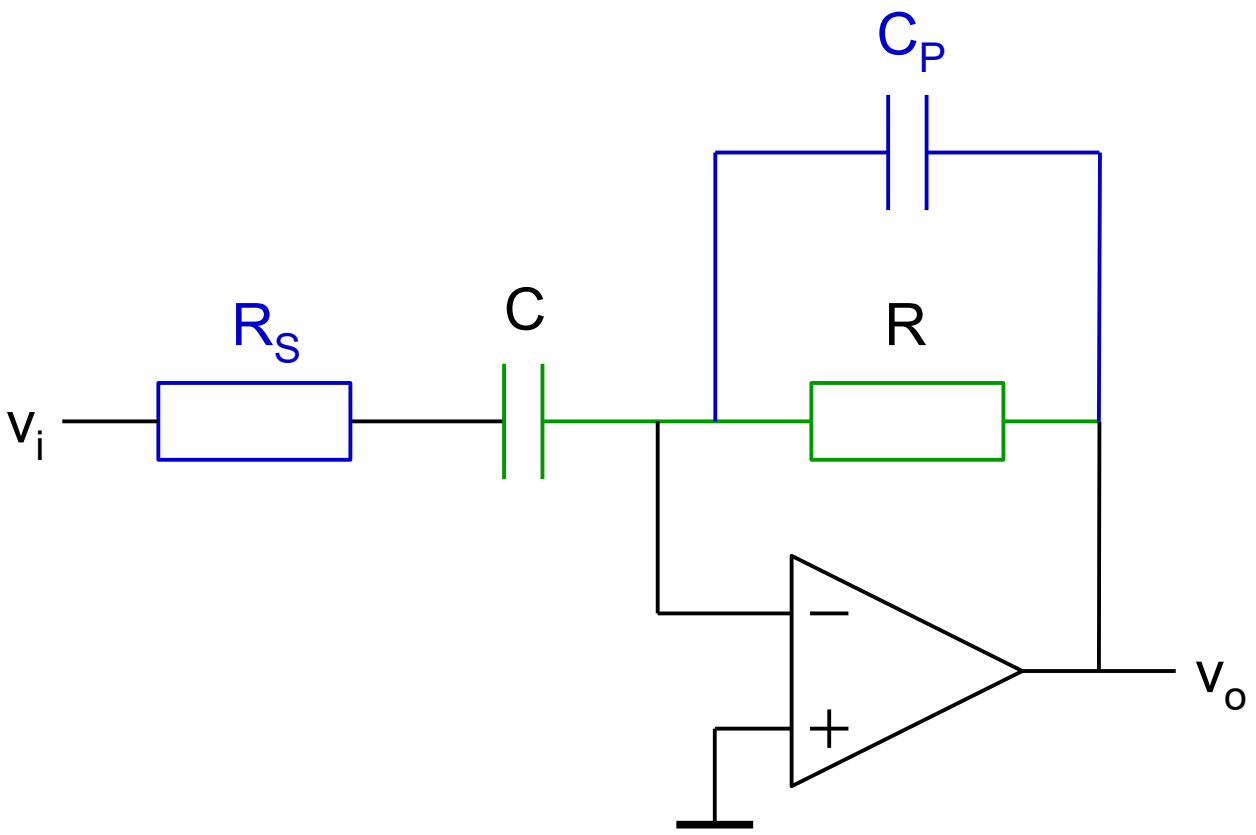
- răspuns la semnal-treaptă
- amplificarea zgomotului de frecvență mare
- stabilitate

Derivator (2)

AO încearcă să mențină 0V.
Circuitul RC îi întârzie reacția.

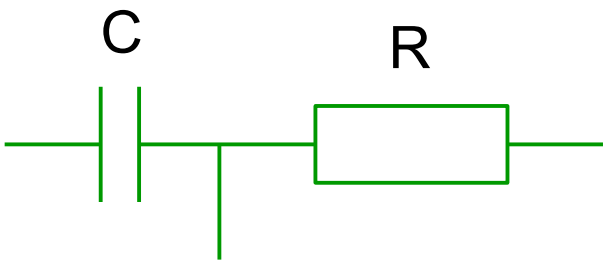


“Derivator” stabilizat

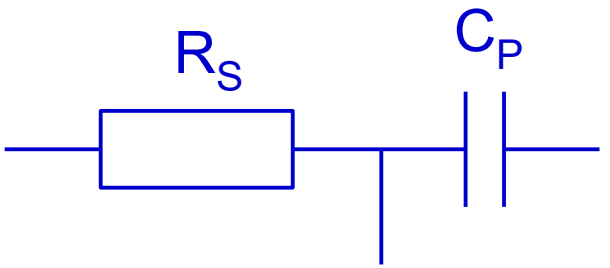


$$R_S \ll R, \quad C_P \ll C$$

la frecvență mică:



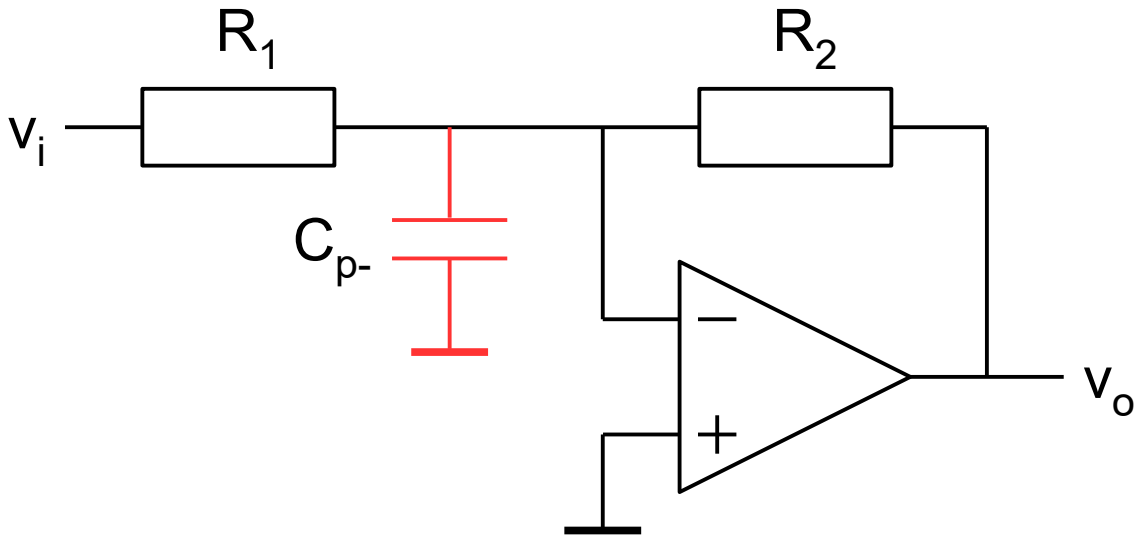
la frecvență mare:

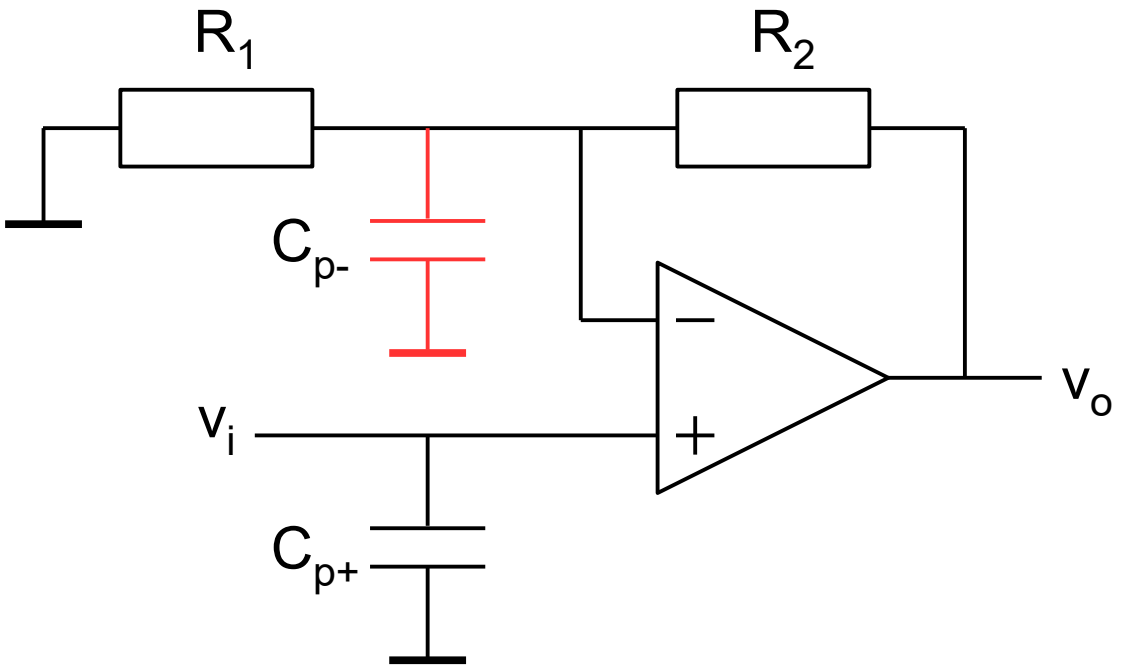


Observație

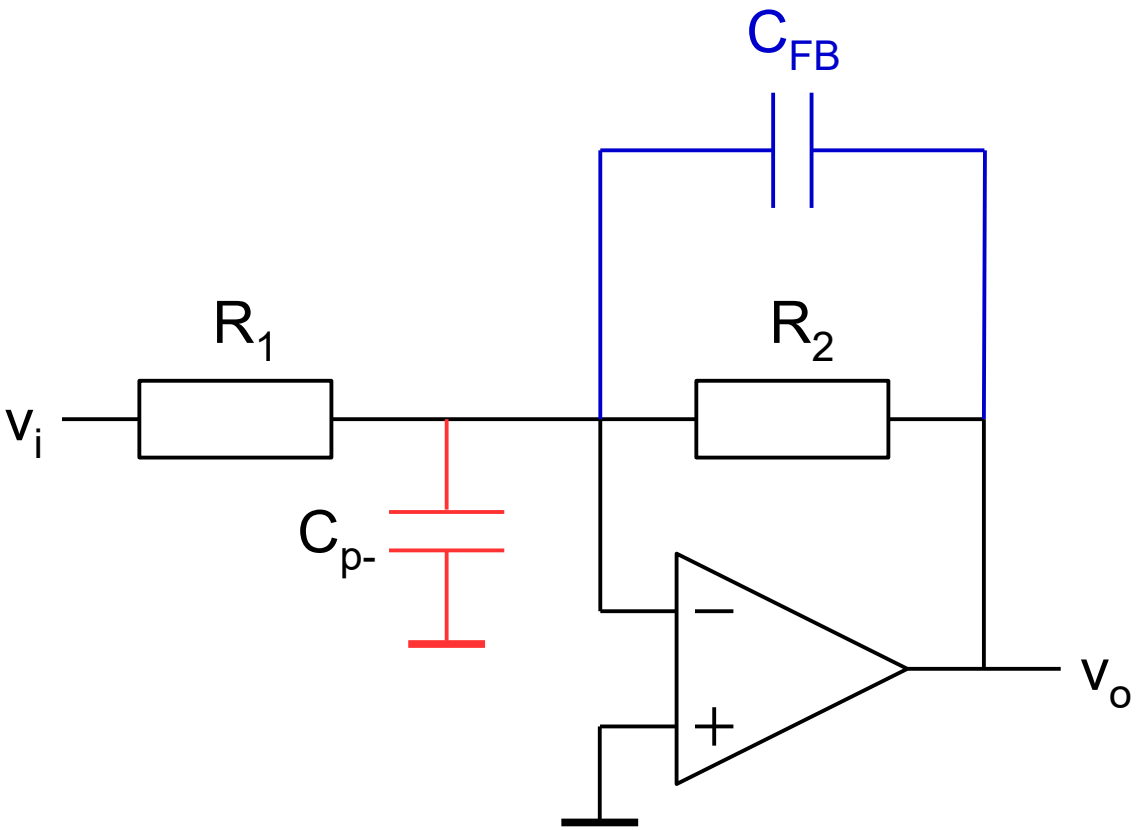
- Și alte tipuri de circuite cu AO pot avea probleme de stabilitate
  - capacitate parazită pe intrarea inversoare
  - capacitate parazită pe ieșire (ex. cablu lung)

Capacitate pe nodul sumator (1)

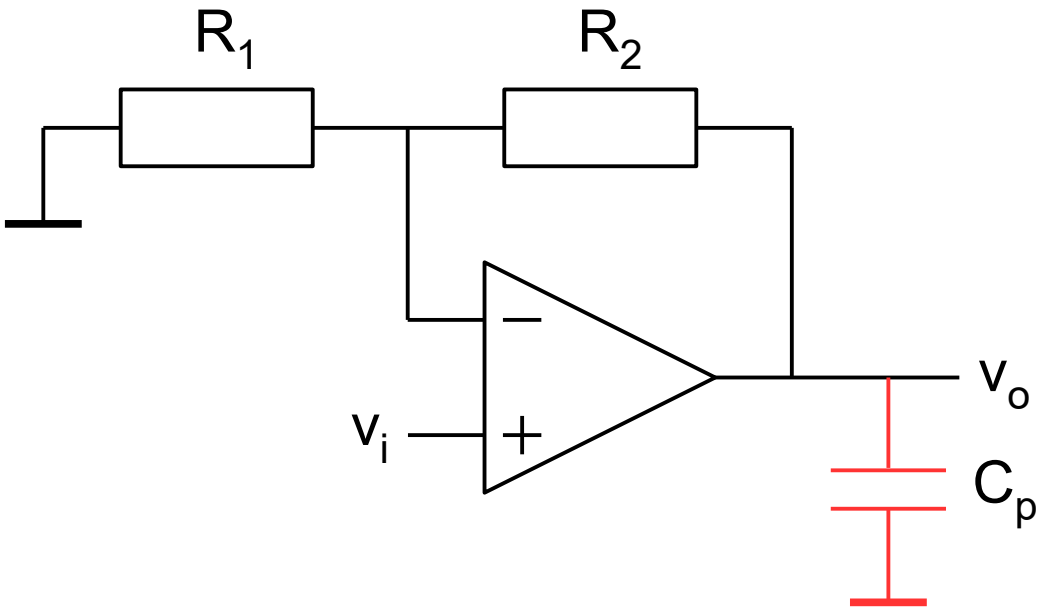




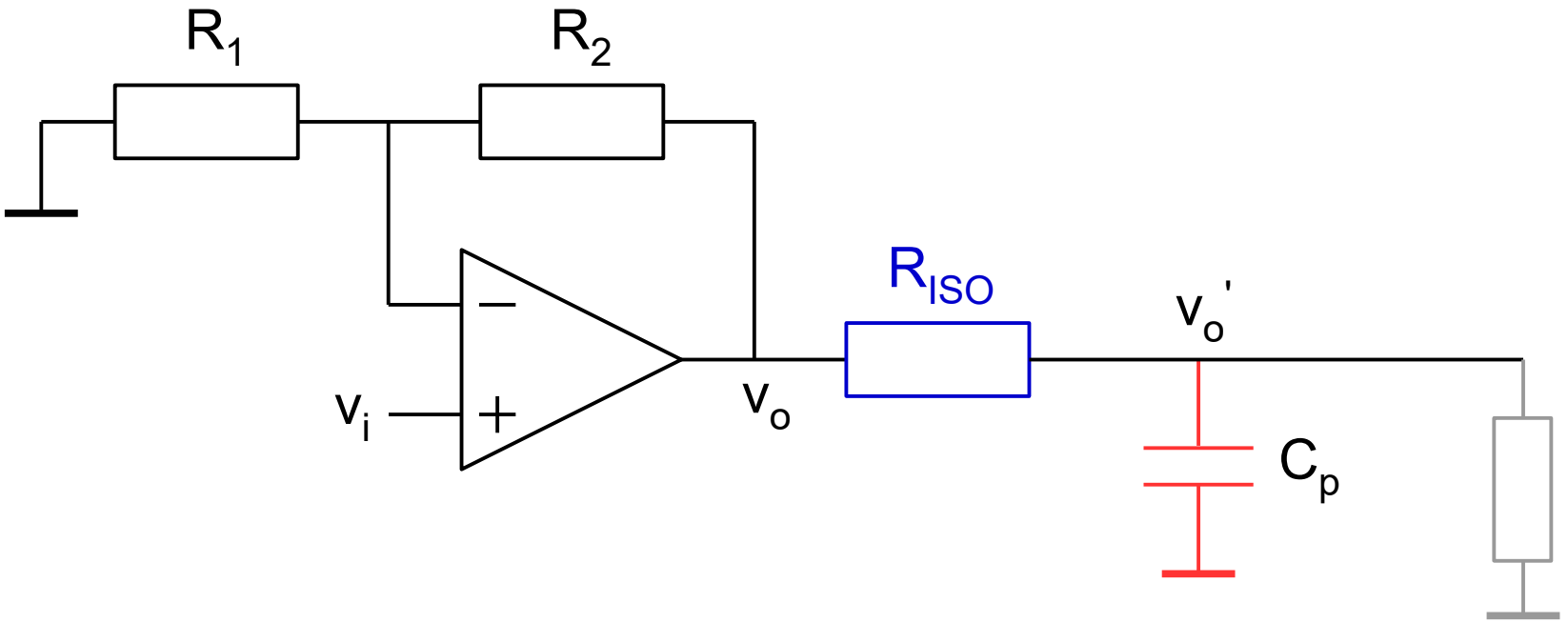
Capacitate pe nodul sumator (2)



Capacitate pe ieșire (1)



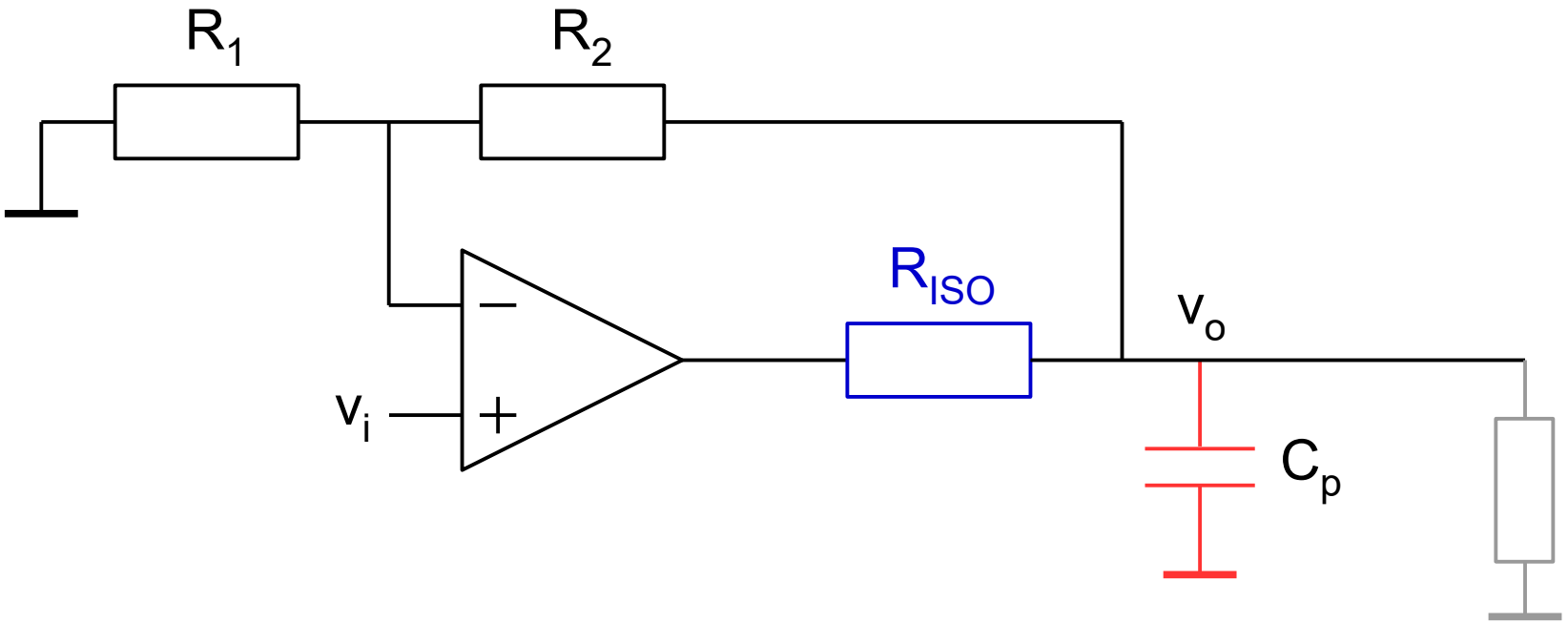
Capacitate pe ieșire (2)



Rezistența serie decuplează (izolează) capacitatea de sarcină de ieșirea AO

problemă: divizor cu rezistența de sarcină

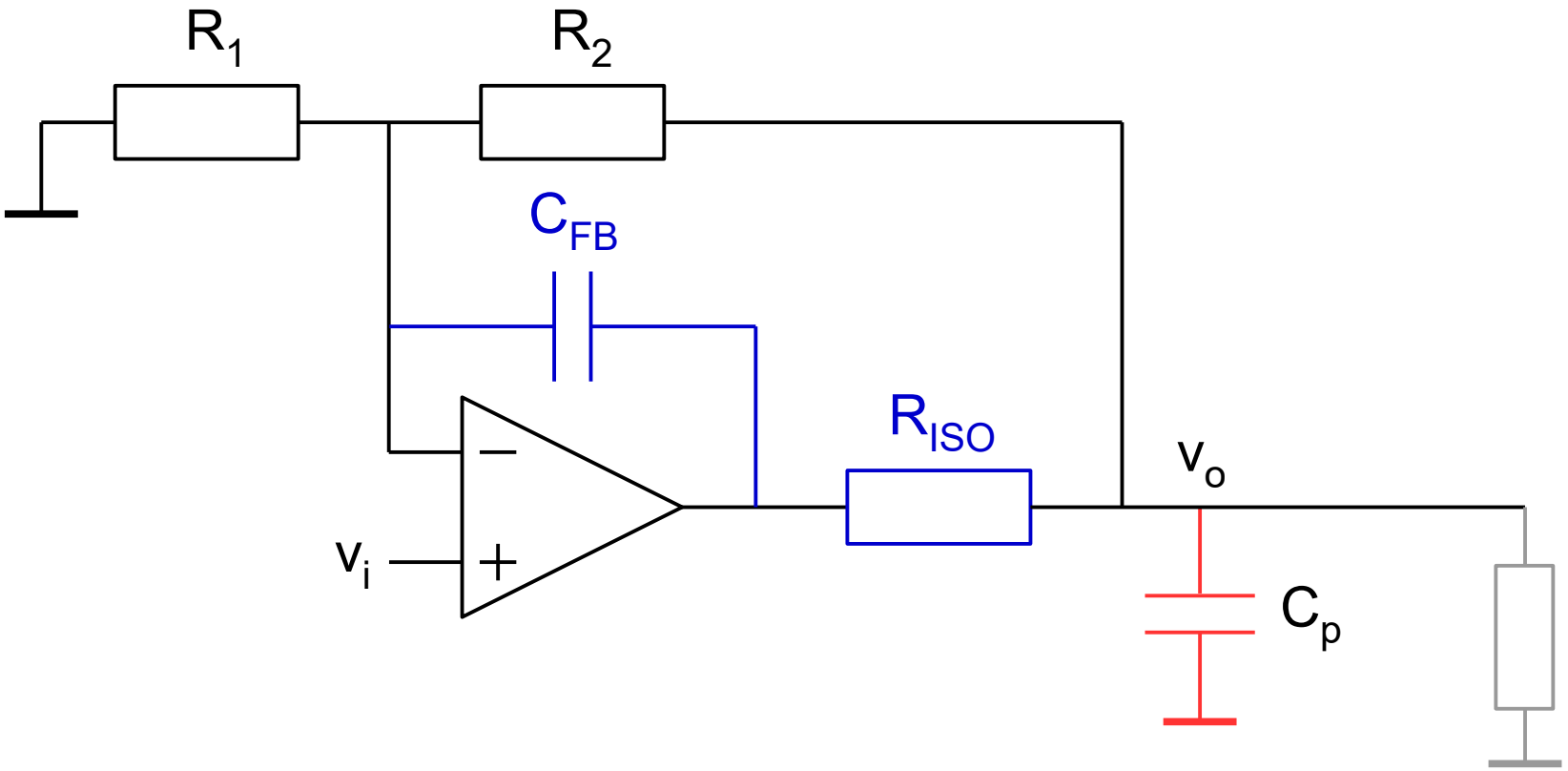
Capacitate pe ieșire (3)



Putem muta rezistența serie înăuntrul buclei de reacție...

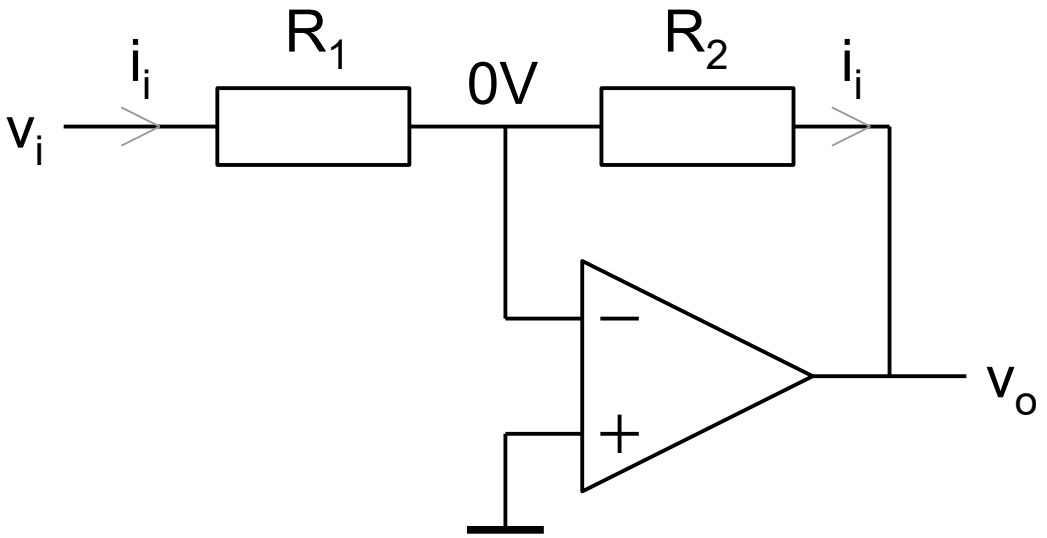
dar am revenit la situația inițială

Capacitate pe ieșire (4)



Introducem o reacție locală rapidă prin C_{FB}

Convertoare V-I, I-V (1)



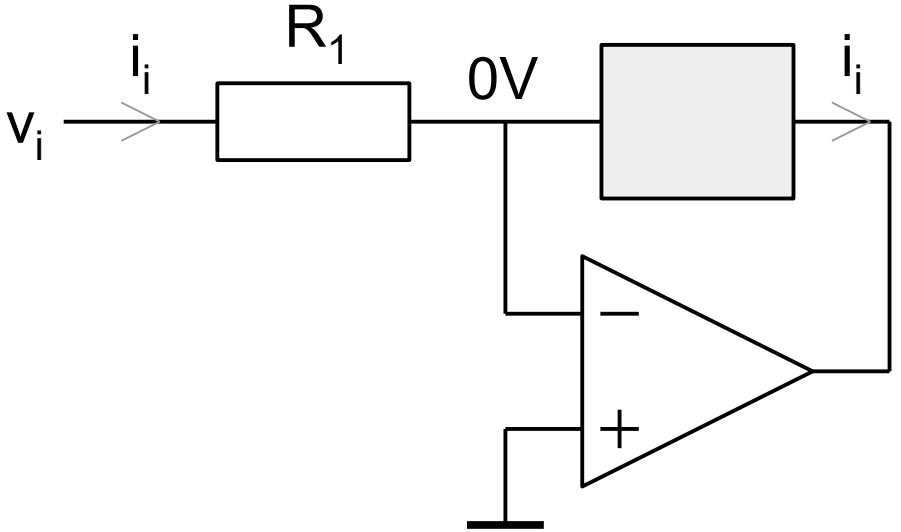
$$i_i = \frac{v_i}{R_1}$$

tensiune => curent

$$v_o = -i_i R_2$$

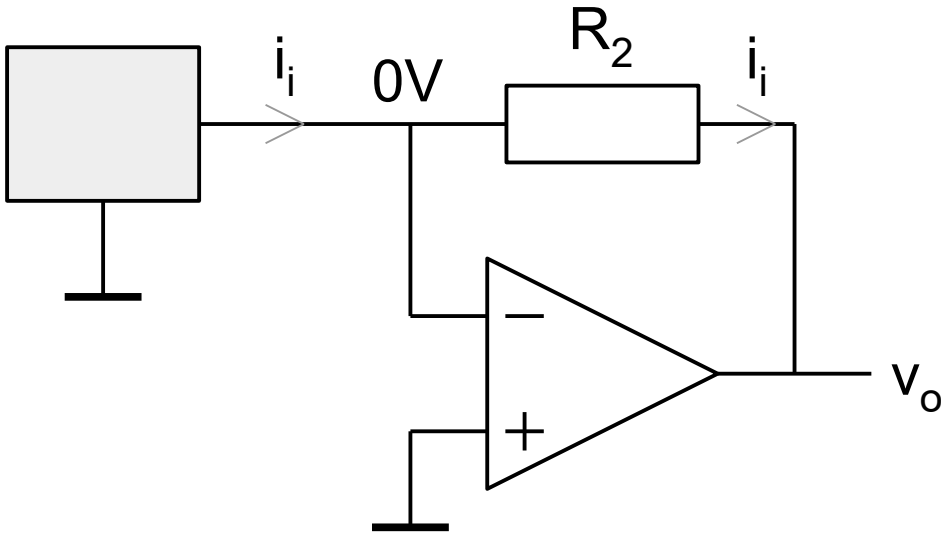
curent => tensiune

Convertoare V-I, I-V (2)



$$i_i = \frac{v_i}{R_1}$$

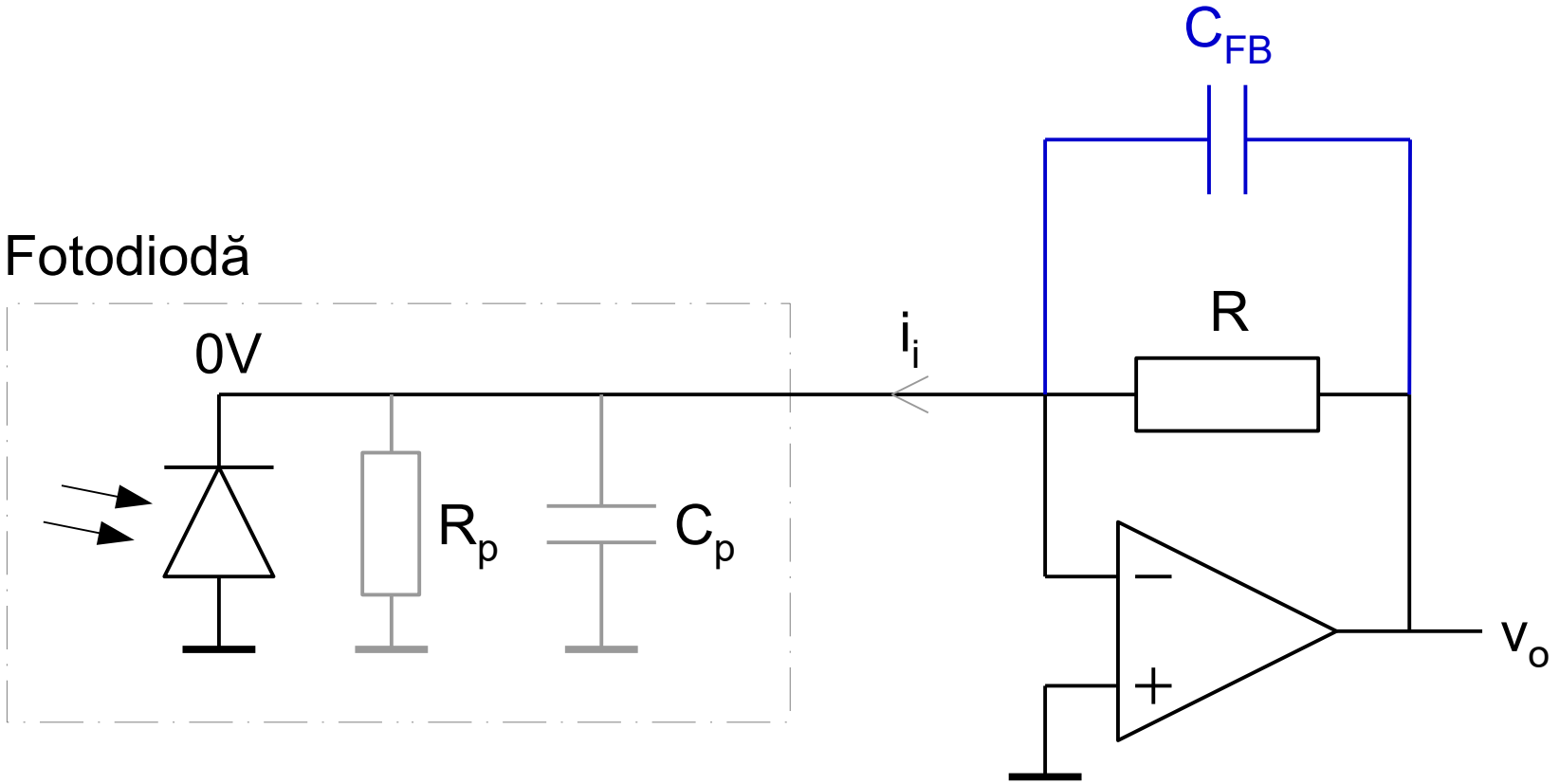
tensiune => curent



$$v_o = -i_i R_2$$

curent => tensiune

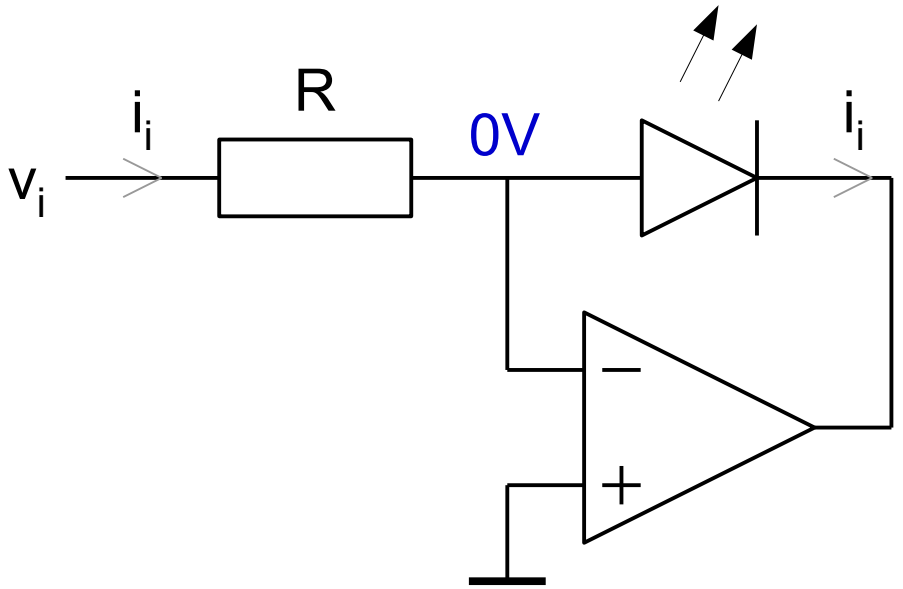
Receptor de semnal optic



fotodioda prezintă:

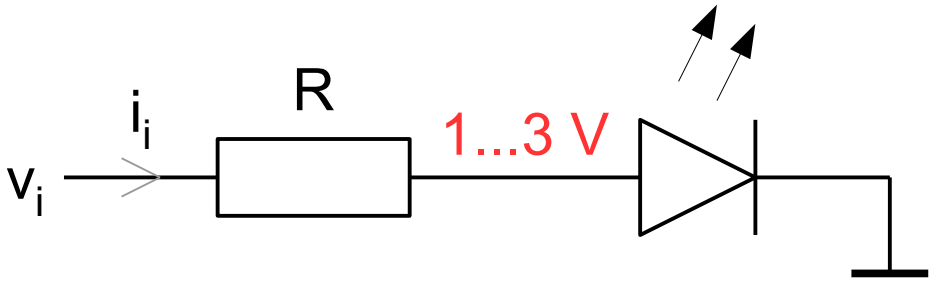
- fotocurent – curent invers, proporțional cu iluminarea
- curent de întuneric (de scurgere)
  - dependent de tensiune (rezistență parazită)
- capacitate parazită

Transmițător de semnal optic



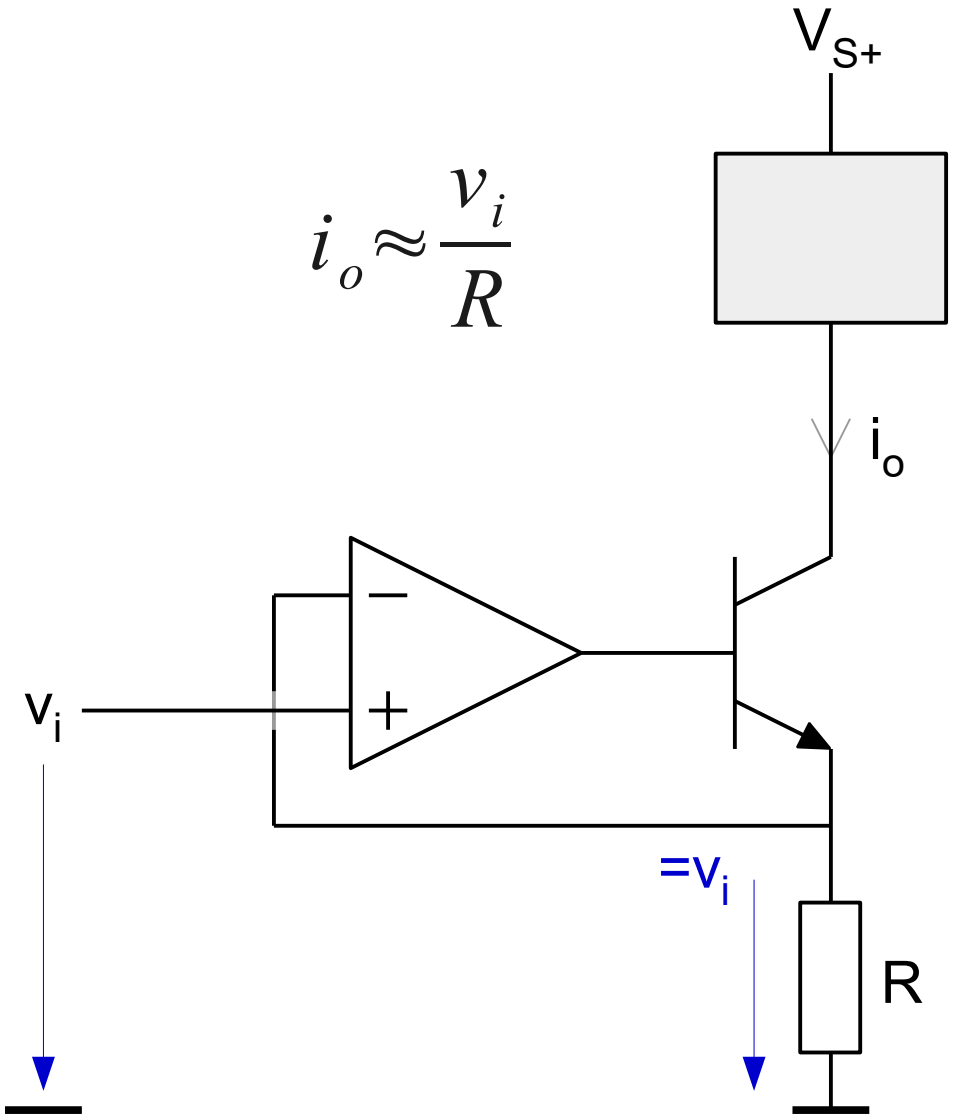
OBS. 1: în general AO tolerează curent de ieșire mic (< 20 mA)

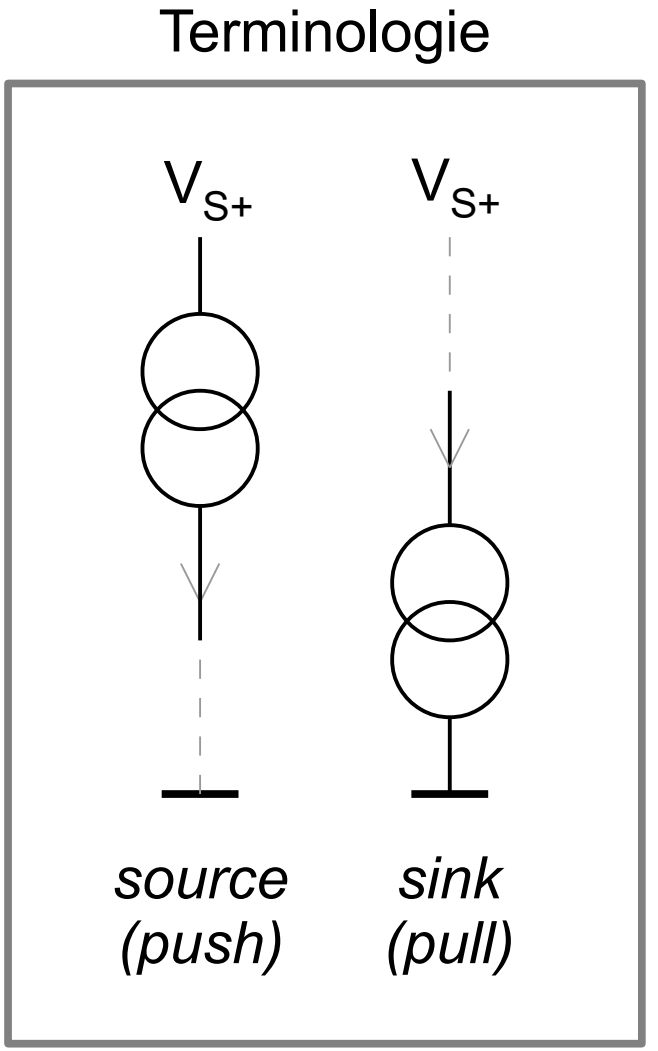
OBS. 2: circuit mai puțin precis:



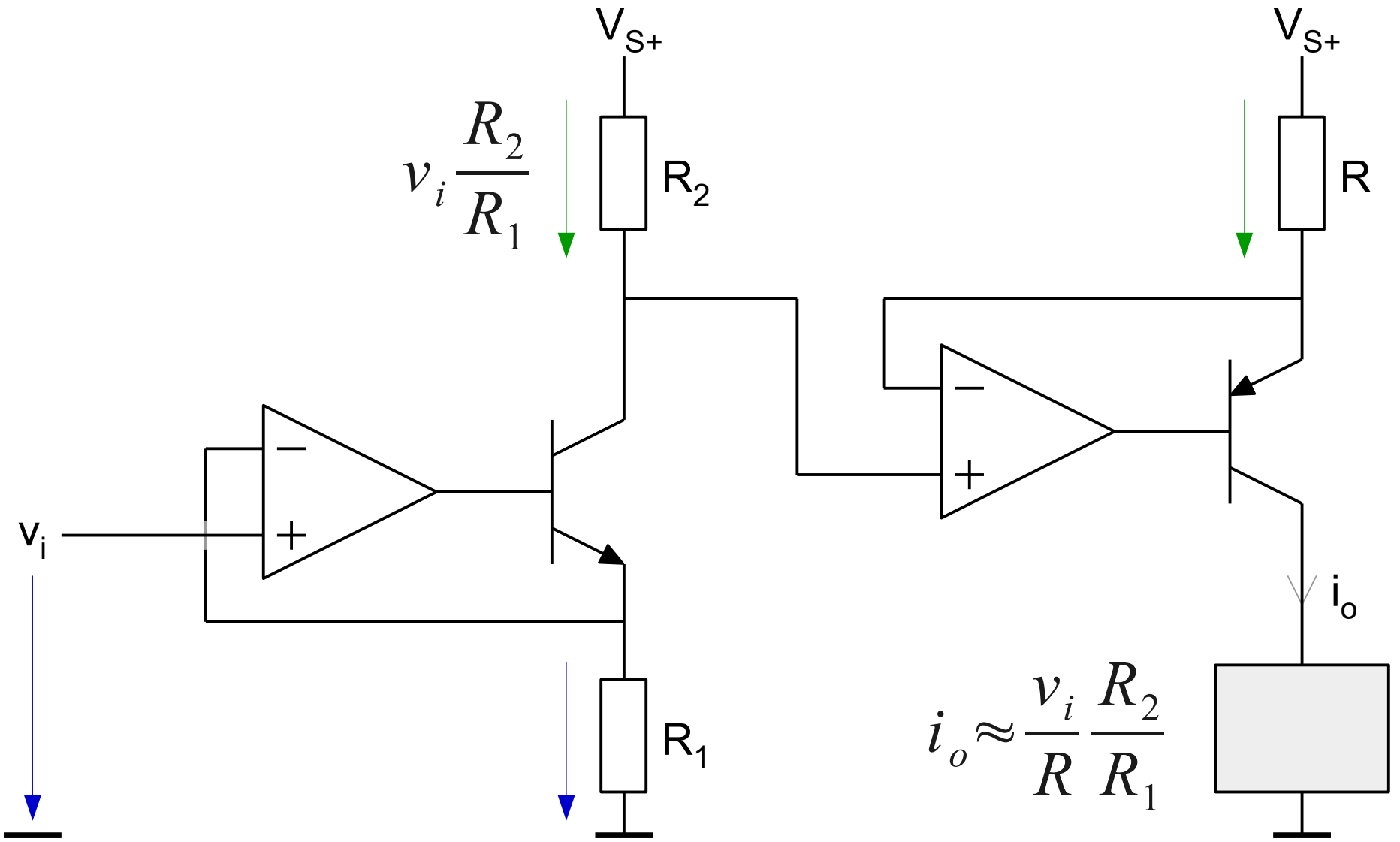
Convertor V-I tip *sink*

$$i_o \approx \frac{v_i}{R}$$





Convertor V-I tip *source*



Amplificator “diferențial”

